D5

RS data covering each of the 57 remaining collection sites

Executive summary

To assist identification of shea rich zones, Sensonomic created a web-based atlas derived from Remote Sensing (RS) data, covering areas of particular relevance and interest. The output is a compilation that can be queried for spatial information of given territories and used to locate areas for further studies. The atlas is made available online.

The atlas consists of commune-level overview true color satellite imagery, false color (vegetation) satellite imagery, land cover / land use (LCLU), fractional tree cover maps, as well as higher resolution true color satellite imagery on a sub-commune level.

This report contains extracts of the full deliverable, and the full product is available on the SeriousShea geographical information system (GIS).

Introduction

In deliverable D-5, Sensonomic produced RS derived data 'covering each of the 57 remaining collection sites'. In preparation for the forestry assessment survey, the need was for recent satellite imagery that could be used to identify shea collection areas in collaboration with collectors. An atlas was created to cover the 50 communes that had shown most promise in earlier analysis (see D-3, among others).

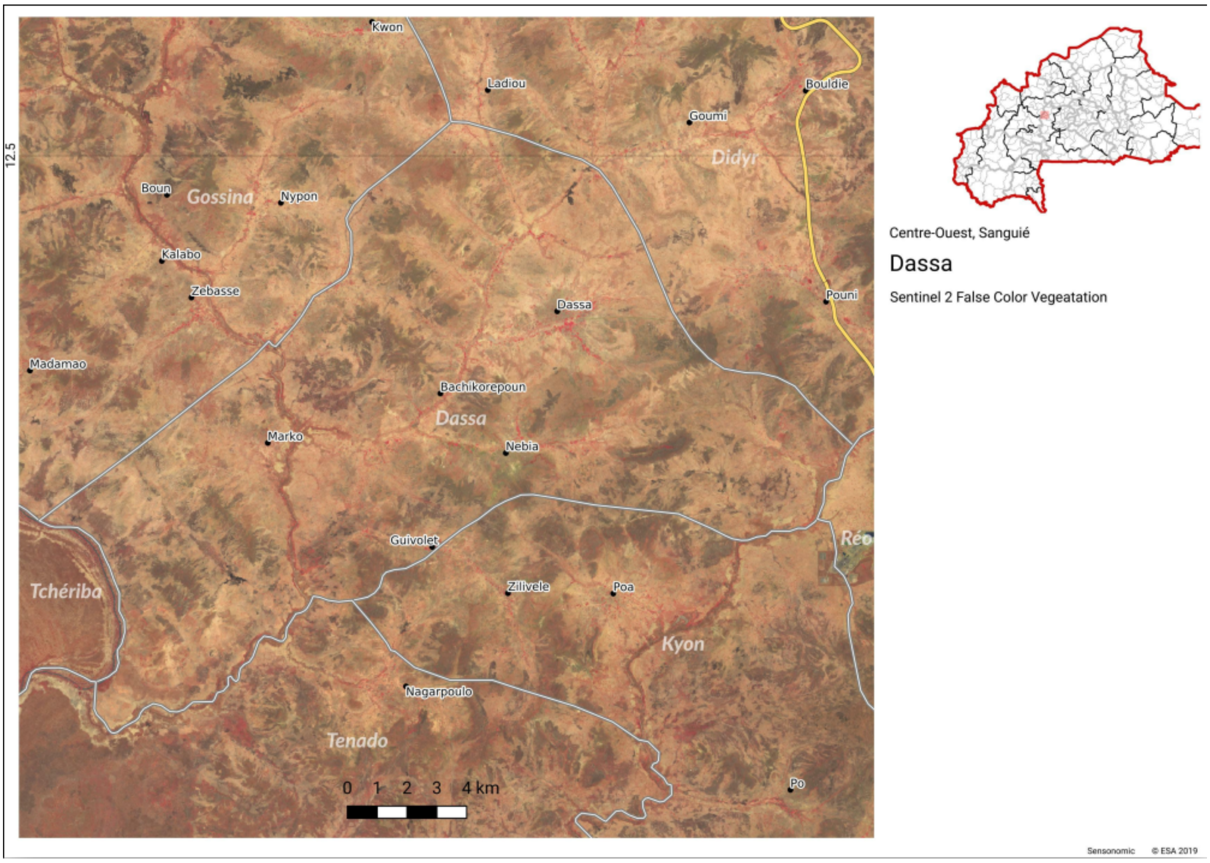
Extracts are reproduced in this report, as the file size of this type of product is beyond what is practical. The full product is available to Serious Shea on our GIS platform, filed under "Serious Shea shared data > Component 3 (Sensonomic) > RS".

The overall goal of the product was to assist in locating shea rich zones. The included layers were selected based on their concerted ability to do so. While true color imagery gives a visually realistic impression of the terrain, false color (vegetation) imagery highlights chlorophyllic activity by displaying it as red. The LCLU layer gives an overview of where to find different types of terrain, like forests, cropland and shrubland, while the fractional tree cover layer indicates the percentage of tree cover per pixel.

Deliverables

True color overview satellite images

European Space Agency Sentinel 2 Level 2A imagery from the period 1st of March 2019 to 1st of June 2019 was sourced from https://scihub.copernicus.eu/. The image collection was then mosaicked based on the "CLOUDY_PIXEL_PERCENTAGE" metadata property to generate a mostly cloud free image of the area of interest. To produce true color imagery, bands 4, 3, and 2 were selected. Sentinel 2 imagery has a resolution of 10 meters per pixel.

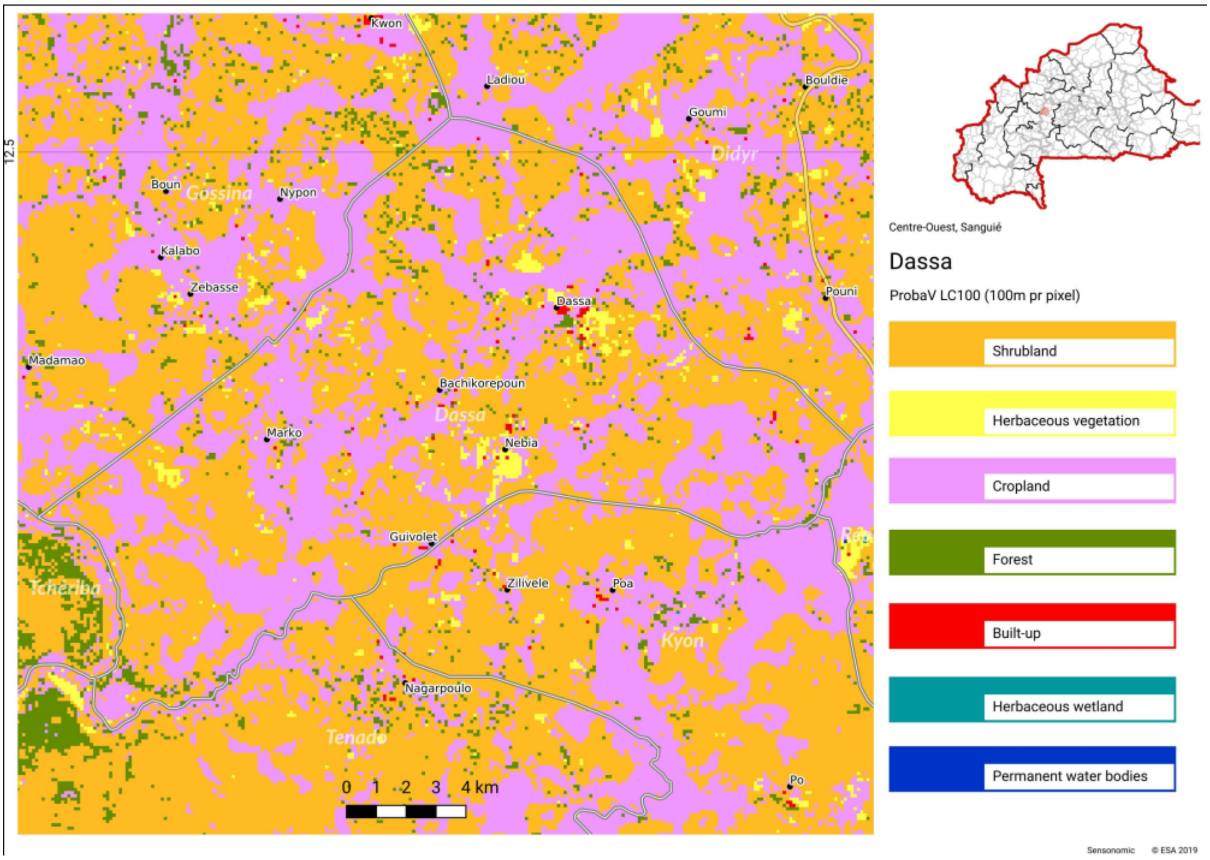


False color (vegetation) overview satellite images

European Space Agency Sentinel 2 Level 2A imagery from the period 1st of March 2019 to 1st of June 2019 was sourced from https://scihub.copernicus.eu/. The image collection was then mosaicked based on the "CLOUDY_PIXEL_PERCENTAGE" metadata property to generate a mostly cloud free image of the area of interest. To produce false color (vegetation) imagery, bands 5, 4, and 3 were selected. Sentinel 2 imagery has a resolution of 10 meters per pixel.

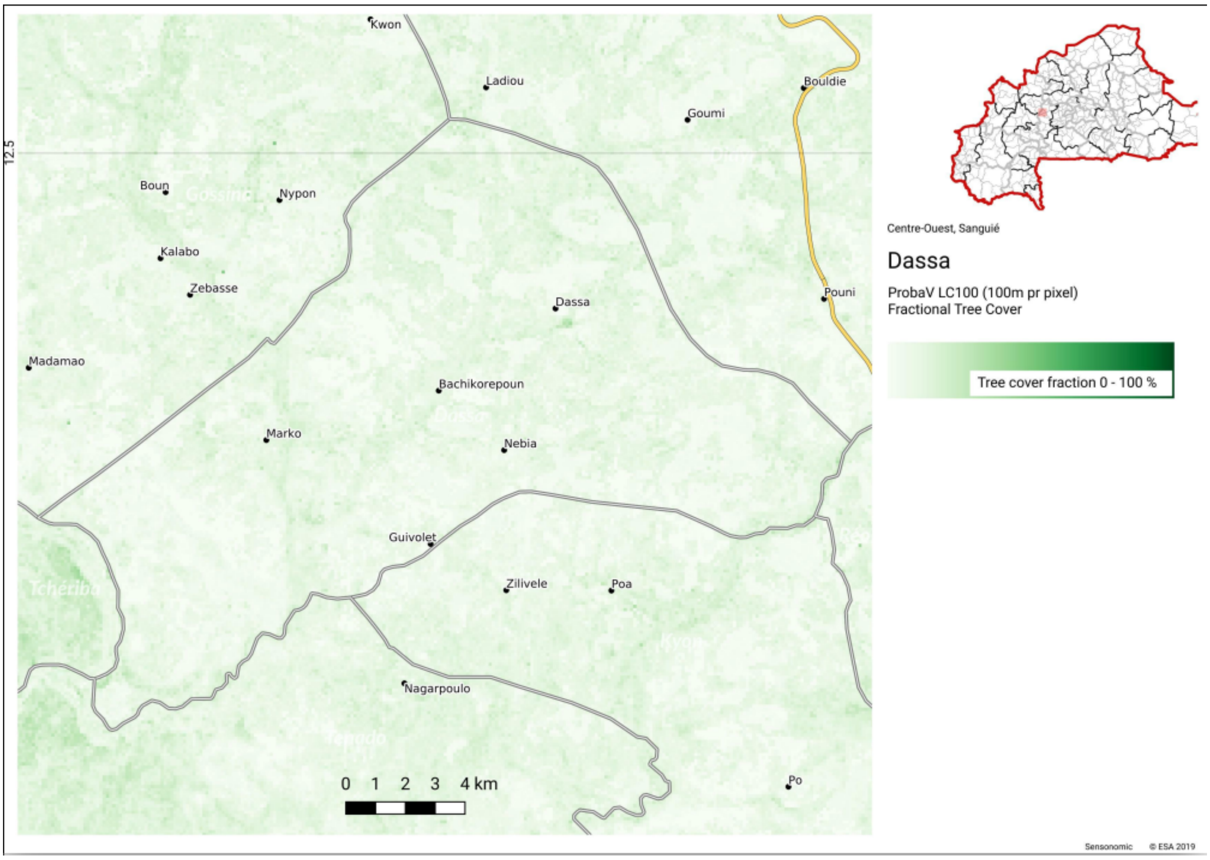
Land cover / land use

The LCLU data used is ESA's Global 100m Land Cover maps for 2015, derived from the PROBA-V satellite, and published on 2019-05-14. The data was pulled from https://lcviewer.vito.be/download. Only the macro classes were utilized as the subclassifications were not useful for our purpose. The data has a resolution of 100 meters per pixel.



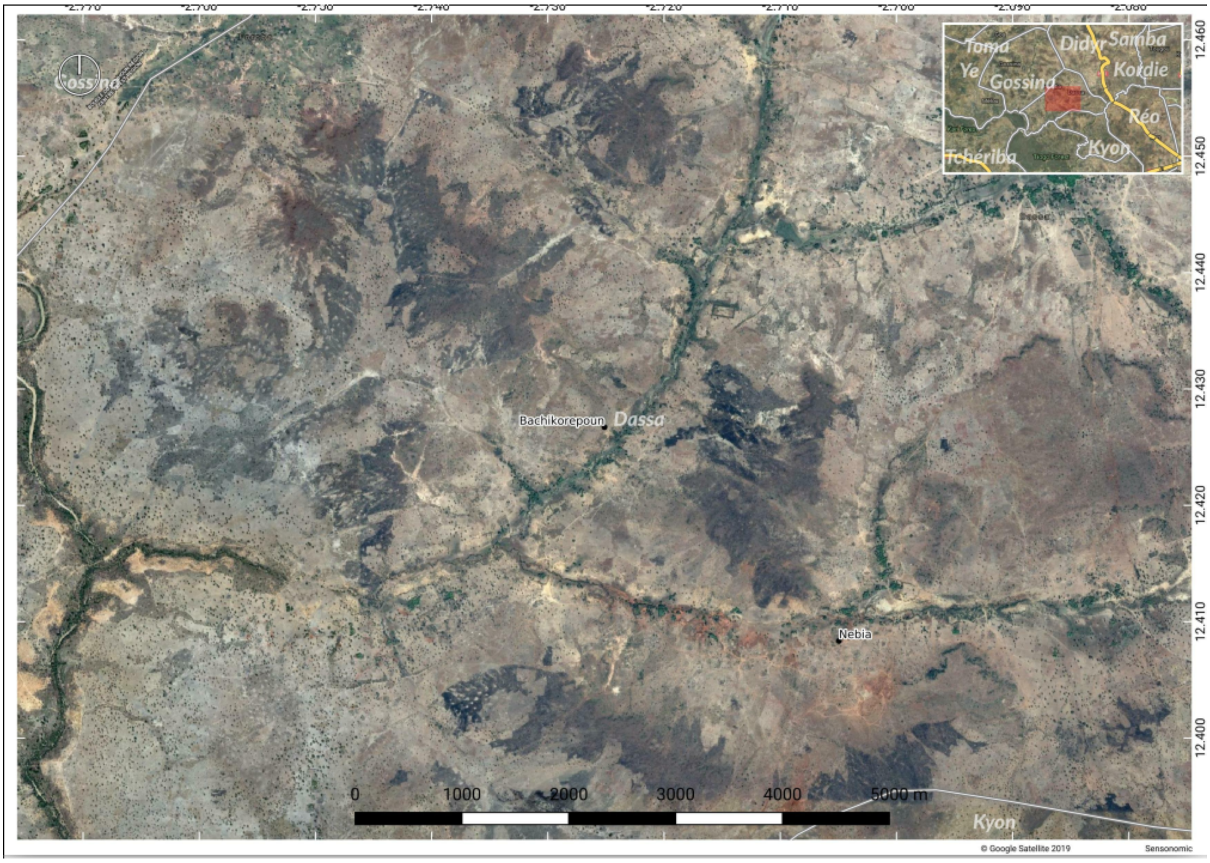
Fractional tree cover

The fractional tree cover data used is another part of the LCLU product described directly above. They share the same capture date and resolution.



Sub commune level satellite images

For sub commune level imagery, Google Satellite Very High Resolution imagery was collected. The Google Satellite product is made up from several sources, but generally has a resolution below 5 meters per pixel. Capture dates vary across space, but for Burkina Faso we found that most of the imagery was captured in 2016.



Methodology

The compilation was done using QGIS, iterating over each of layers for each of the communes. Each commune was covered by about ten pages. To create the sub commune level products, a 10x10km grid was produced, and for the relevant communes iterated over, to reveal more detail. Place names, commune borders and road network layers were added on top to make the atlas easier to read.

Conclusion

50 communes of particular interest were extensively mapped, totalling over 600 pages, offering information on terrain types, tree cover, and more to Serious Shea. This geographically explicit information will assist in selection and ranking of communes for the next stages in the SeriousShea project.